Rasel Ahmed Bhuiyan

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Research Interests

Computer Vision, Machine Learning, Human Biometrics, Biometric Security, Foundation and Vision-Language Models, Generative AI, Explainable AI, Biomedical and Medical Image Analysis

Education

Ph.D. Computer Science & Eng. – University of Notre Dame, Notre Dame, IN — Expected by May 2027
Focus: Iris Recognition at Life Extremes (Infant and Post-Mortem) | Advisor: Adam Czajka

M.Sc. Computer Science & Engineering – University of Notre Dame, Notre Dame, IN — 08/2025
Focus: Forensic Iris Recognition | CGPA: 3.83/4.00

B.Sc. Computer Science & Engineering – University of Asia Pacific, Dhaka, BD — 03/2018
Graduated with **highest distinction** | CGPA: 3.94/4.00

Awards & Honors

- **Best Paper Award** – IEEE/CVF WACV 2025 Workshop (CV4Smalls) for the infant iris recognition system.
- **Vice Chancellor's Award** – Six consecutive semesters of academic distinction during B.Sc., University of Asia Pacific.

Research Support

- **National Science Foundation (NSF), Grant 2237880** – Infant Biometric Identification for Healthcare Linkage. Lead graduate researcher on the infant iris recognition work, from data curation to publication.

Research Experience

Graduate Research Assistant — 01/2022 – Present
Computer Vision Research Laboratory (CVRL), University of Notre Dame — *Notre Dame, IN*

- **Infant Iris Recognition (NSF Funded):** Led the work from data curation to publication. Curated and quality-controlled near-infrared iris images of infants aged four to six weeks, and evaluated adult iris recognition systems on them; some commercial systems rejected every image because their segmentation failed. Trained a Nested Residual U-Net segmentation model without any labeled infant images, using infant-specific augmentations of adult iris samples, and built the first operational infant iris recognition system (EER ≈ 3%, AUC 0.99 in a clinical population), with applications to preventing baby swapping, reducing abductions, and improving mother–infant linkage. Released a privacy-preserving synthetic infant iris dataset. GitHub | Related paper: [WACVW] (**Best Paper Award**)
- **Post-Mortem Iris Recognition and Forensic Examination:** Developed a set of computer science tools that help criminal investigators analyze post-mortem iris samples, including post-mortem iris segmentation and matching. Analyzed one of the largest post-mortem iris datasets to date (5,770 near-infrared and 4,643 visible-light images from 259 deceased subjects, with post-mortem intervals up to 1,674 hours) and ran a demographic analysis across more than 32 million comparisons, which showed no consistent effect of gender and higher accuracy for middle-aged subjects. Related papers: [T-BIOM] [JFI survey]
- **ColdIrisNet — Foundation-Model Adaptation for Forensic Recognition:** Adapted a frozen DINOv3 (ViT-L/16) foundation model using ArcFace metric learning and a saliency-guided multi-task loss, paired with a custom landmark-localization network that keeps features aligned under rotation of up to ±80°. Reduced Equal Error Rate by 68% over prior state-of-the-art post-mortem iris matchers (EER ≈ 2.9%) across 70 days of post-mortem change, and demonstrated robust cross-dataset and live-to-deceased transfer performance. Related paper: [Information Fusion, Elsevier, under review]
- **Post-Mortem Interval (PMI) Estimation:** Developed the first method to estimate PMI directly from iris images. Benchmarked deep regression architectures and built a model that fuses near-infrared and visible-light images (MAE of 3.4 hours in sample-disjoint and 17.5 hours in subject-disjoint evaluation). Balancing the training data with synthetic PMI-conditioned images reduced cross-laboratory error by 40% on data acquired in the U.S. and in Poland. Current work adapts DINOv3 with frozen, partially unfrozen, and LoRA strategies and attention-based multiple instance learning to estimate PMI from a single iris image, reducing MAE by 31.4% compared with state-of-the-art iris-based estimators (paper in preparation). Built subject-disjoint data split tools, and am extending saliency-guided training to this task using expert annotations. GitHub | Related paper: [WACV]

- **Multimodal LLMs and Vision-Language Models for Forensic Estimation:** Developed a retrieval-augmented, few-shot framework in which multimodal LLMs (LLaMA-3.2-Vision, Qwen-3-VL) estimate PMI from iris images, using MPNet-based semantic retrieval to calibrate prompts and select few-shot examples; benchmarked against ResNet-152 and foundation-model regression baselines. Now training SigLIP- and BioCLIP-based image–text models that estimate PMI from a near-infrared iris image and its expert visual description, comparing frozen encoders, last-layer fine-tuning, and LoRA.
- **Synthetic Data Generation:** Designed a first-of-its-kind PMI-conditioned generative model (StyleGAN2-ADA) for forensic iris images, with latent-space hypersphere sampling for controllable identity synthesis. Generated over 180K images across 18 PMI ranges that pass 11 of 12 ISO/IEC 29794-6 iris image quality checks, for training data augmentation and forensic examiner training without additional cadaver samples. GitHub | Related paper: [WACVW]
- **Iris Presentation Attack Detection:** Trained presentation attack detection models on more than 400,000 images covering live eyes and many attack types (printouts, synthetic irises, Kindle displays, textured contact lenses, post-mortem eyes, and artificial eyes), reaching over 95% accuracy on attack types held out from training. Built the training pipeline with crop caching and multi-GPU training.
- **Iris Toolbox:** Developed and delivered a foundation model-based iris image processing toolbox to the NIST IREX-vetted OpenSourceIrisRecognition toolkit, comprising models for iris boundary estimation, eye-canthi localization and alignment, eyelid curve fitting, and projective gaze correction. The canthi detection model now runs in NIST's internal iris recognition pipeline. GitHub
- **Visualizing Brain-Stimulation-Induced Perceptual Events:** Designed generative computer vision models (GANs and diffusion models) for in-manifold and out-manifold latent space exploration for a novel neuroscience tool called *perceptographer*, which visualizes complex perceptual distortions induced by local stimulation of the inferotemporal cortex. Co-authored a follow-up study that introduces porosity, a measure of how completely an image-generation engine can span a high-dimensional stimulus space. Collaboration with the National Institute of Mental Health, USA (NIMH). Related paper: [JOV abstract]

Research Lead (as Lecturer) 10/2018 – 12/2021
Uttara University *Dhaka, Bangladesh*

- **Epileptic Seizure Recognition:** Statistical analysis of EEG signals along with designing and implementing classification strategies for automated seizure diagnosis. Related paper: [ICIEV]
- **Human Activity Recognition:** Designed and implemented frequency-domain models introducing improved feature extraction, dimensionality reduction, and dictionary-based signal representations for wearable patient monitoring. Related papers: [SIVP] [Sensors] [ICKII]

Research Assistant 03/2016 – 09/2018
Computer Vision & Pattern Recognition Lab, University of Asia Pacific *Dhaka, Bangladesh*

- **Hand Gesture Recognition:** Developed a gesture recognition system utilizing genetic algorithm–based optimization, enabling enhanced communication for deaf and hard-of-hearing users. Related paper: [ICCIT]
- **American Sign Language Recognition:** Built a low-cost, real-time system using deep learning-based feature extraction to support interactive communication for deaf and hard-of-hearing users. Related paper: [ICFSP]
- **Arrhythmia Diagnosis:** Designed and implemented an ECG-based method for arrhythmia classification. Related paper: [HNICEM]

Publications

14 peer-reviewed publications (6 journal, 6 conference, 2 workshop), plus one manuscript under review and one submitted.
Google Scholar: https://scholar.google.com/citations?user=LvUO52oAAAAJ&hl=en

Manuscripts Under Review and Submitted:

R1. **Rasel Ahmed Bhuiyan** and Adam Czajka, "ColdIrisNet: Biology-Guided Fusion for Forensic Iris Recognition," *Information Fusion, Elsevier*, 2026. (Under Review)

R2. Elia Shahbazi, Drew Nguyen, **Rasel Ahmed Bhuiyan**, Adam Czajka, Arash Afraz, "Perceptographer: AI-Powered Framework for Controlled Navigation of High-Dimensional Stimulus Spaces," manuscript submitted to *Journal of Vision*, 2026.

Journal Publications:

J1. **Rasel Ahmed Bhuiyan**, Parisa Farmanifard, Renu Sharma, Andrey Kuehlkamp, Aidan Boyd, Patrick J. Flynn, Kevin W. Bowyer, Arun Ross, Dennis Chute, and Adam Czajka, "Beyond Mortality: Advancements in Post-Mortem Iris Recognition through Data Collection and Computer-Aided Forensic Examination," *IEEE Transactions on Biometrics, Behavior, and Identity Science (T-BIOM)*, vol. 7, no. 4, pp. 808–823, 2025.

J2. Elia Shahbazi, Drew Nguyen, **Rasel Ahmed Bhuiyan**, Adam Czajka, Arash Afraz, "Visualizing the Unseen: Perceptographer, an AI Engine for Visualizing Brain-Stimulation-Induced Perceptual Events," *Journal of Vision*, vol. 25, p. 2855, July 2025. (Abstract)

J3. **Rasel Ahmed Bhuiyan** and Adam Czajka, "Forensic Iris Recognition: A Survey," *Journal of Forensic Identification*, vol. 74, no. 1, p. 38, 2024.

J4. **Rasel Ahmed Bhuiyan**, Shams Tarek, and Hongda Tian, "Enhanced Bag-of-Words Representation for Human Activity Recognition using Mobile Sensor Data," *Signal, Image and Video Processing (SIViP)*, Springer Nature, pp. 1–8, 2021.

J5. **Rasel Ahmed Bhuiyan**, A.K. Kundu, S.R. Shafi, and A. Matin, "A Bag-of-Words Based Feature Extraction Scheme for American Sign Language Number Recognition from Hand Gesture Images," *International Journal of Machine Learning and Computing (IJMLC)*, vol. 11, no. 1, pp. 85–91, 2021.

J6. **Rasel Ahmed Bhuiyan**, N. Ahmed, Md. Amiruzzaman, and M.R. Islam, "A Robust Feature Extraction Model for Human Activity Characterization using 3-Axis Accelerometer and Gyroscope Data," *Sensors*, MDPI, vol. 20, no. 23, Art. no. 6990, 2020.

Conference Publications:

C1. **Rasel Ahmed Bhuiyan** and Adam Czajka, "Forensic Iris Image-Based Post-Mortem Interval Estimation," *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Tucson, Arizona, 2025, pp. 4258–4267.

C2. **Rasel Ahmed Bhuiyan**, Md. Amiruzzaman, Nadeem Ahmed, Md. Rashedul Islam, "Efficient Frequency Domain Feature Extraction Model using EPS and LDA for Human Activity Recognition," *3rd IEEE International Conference on Knowledge Innovation and Invention (ICKII)*, Kaohsiung, Taiwan, 2020, pp. 344–347.

C3. A. Matin, **Rasel Ahmed Bhuiyan**, S.R. Shafi, A.K. Kundu, and M.U. Islam, "A Hybrid Scheme Using PCA and ICA Based Statistical Feature for Epileptic Seizure Recognition from EEG Signal," *Joint 2019 IEEE 8th International Conference on Informatics, Electronics, and Vision (ICIEV) and 3rd International Conference on Imaging, Vision, and Pattern Recognition (IVPR)*, Eastern Washington University, USA, 2019.

C4. Md. Rashedul Islam, **Rasel Ahmed Bhuiyan**, Nadeem Ahmed, Md. Rajibul Islam, "PCA and ICA Based Hybrid Dimension Reduction Model for Cardiac Arrhythmia Disease Diagnosis," *2018 IEEE 10th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment and Management (HNICEM)*, Baguio City, Philippines, 2018, pp. 1–7.

C5. Md. Rashedul Islam, Ummey Kulsum Mitu, **Rasel Ahmed Bhuiyan**, Jungpil Shin, "Hand Gesture Feature Extraction Using Deep Convolutional Neural Network for Recognizing American Sign Language," *4th International Conference on Frontiers of Signal Processing (ICFSP)*, Poitiers, France, 2018, pp. 115–119.

C6. **Rasel Ahmed Bhuiyan**, Abdul Kawsar Tushar, Akm Ashiquzzaman, Jungpil Shin, Md. Rashedul Islam, "Reduction of Gesture Feature Dimension for Improving the Hand Gesture Recognition Performance of Numerical Sign Language," *20th International Conference of Computer and Information Technology (ICCIT)*, Dhaka, Bangladesh, 2017, pp. 1–6.

Workshop Publications:

W1. **Rasel Ahmed Bhuiyan**, Mateusz Trokielewicz, Piotr Maciejewicz, Sherri Bucher, Adam Czajka, "Iris Recognition for Infants," *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops*, 2025, pp. 83–92. **(Best Paper Award)**

W2. **Rasel Ahmed Bhuiyan**, Adam Czajka, "Forensic Iris Image Synthesis," *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops*, 2024, pp. 1015–1023.

Teaching Experience

Graduate Teaching Assistant 01/2022 – Present

Department of Computer Science and Engineering, University of Notre Dame *Notre Dame, IN*

- Contributed to course development, including designing assignments, labs, and project materials.
- Conducted office hours and facilitated class discussions to support student learning.
- Assessed student work, providing constructive feedback to enhance understanding and performance.
- Courses Assisted:
  - CSE 30332: Programming Paradigms (Spring 2022)
  - CSE 20312: Data Structures (Fall 2022)
  - CSE 40868: Neural Networks (Spring 2023, Spring 2024)
  - CSE 40535: Special Studies: Computer Vision (Fall 2023, Fall 2024)
  - CSE 60868: Graduate Neural Networks (Spring 2025)
  - CSE 60535: Graduate Computer Vision (Fall 2025)

Lecturer 10/2018 – 12/2021

Department of Computer Science and Engineering, Uttara University *Dhaka, Bangladesh*

- Served as the primary instructor for undergraduate engineering courses including:
  - Algorithms
  - Digital Logic Design
  - Machine Learning
  - Artificial Intelligence
- Actively designed course materials, assignments, and exam questions to support student learning outcomes.
- Taught courses online over Zoom during the COVID-19 period, in addition to in-person teaching.

Mentoring & Student Supervision

- Mentored two undergraduate students at Notre Dame (Summer and Fall 2023) in building an iris presentation attack detection dataset, with weekly meetings on iris recognition fundamentals and presentation attack detection.
- Supervised undergraduate final-year projects at Uttara University (2018–2021), providing guidance on project development and research methodology.

Talks & Presentations

- Invited Talk, Voxel51, 2025.
- Paper Presenter, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Tucson, Arizona, 2025.
- Paper Presenter, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Waikoloa Beach Marriott Resort, Waikoloa, Hawaii, 2024.
- Paper Presenter, IEEE International Conference of Computer and Information Technology (ICCIT), University of Asia Pacific, Dhaka, Bangladesh, 2017.

Professional Service & Outreach

- **Reviewer:** IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2027 and 2025; International Conference on Science and Contemporary Technologies (ICSCT), 2021.
- **Departmental Service, Uttara University (2018–2021):** Routine Committee, Accreditation Committee, and Faculty Recruitment Committee.
- **Outreach:** Invited to represent the Computer Vision Research Lab at the annual CSE Lab Day, University of Notre Dame (Fall 2025), introducing first-year students to computer vision research with live demonstrations.

Research Collaborations

- **U.S. national institutions:** National Institute of Standards and Technology (NIST) – iris image processing toolbox, with the canthi detection model in NIST's internal pipeline; National Institute of Mental Health (NIMH) – Perceptographer.
- **Academic and other partners:** Indiana University Indianapolis, Michigan State University, Warsaw University of Technology, Medical University of Warsaw, Dutchess County Medical Examiner's Office (New York), and PayEye.

Open-Source Software & Data

- **Infant Iris Recognition:** segmentation and recognition code, and a privacy-preserving synthetic infant iris dataset. GitHub
- **PMI Estimation:** code for post-mortem interval estimation from iris images. GitHub
- **Forensic Iris Synthesis:** code for PMI-conditioned iris image synthesis. GitHub
- **OpenSourceIrisRecognition (contributor):** iris boundary estimation, eye-canthi localization and alignment, eyelid curve fitting, and projective gaze correction modules. GitHub

Technical Skills

- **Core Research:** Computer Vision, Biometrics, Foundation Models (DINOv3) and Parameter-Efficient Fine-Tuning (LoRA), Vision Transformers (ViT), Metric Learning (ArcFace), Multiple Instance Learning, Generative AI (GANs/Diffusion), Vision-Language Models and Multimodal LLMs, Segmentation, Classification, Regression, Object Detection, Domain Adaptation, Explainable AI (XAI).
- **Languages:** Python, MATLAB, Bash, SQL.
- **ML Frameworks:** PyTorch, TensorFlow, Hugging Face, Scikit-learn, OpenCV, NumPy, Pandas.
- **Development & Infrastructure:** Git, Linux, Docker, CUDA, HPC Clusters (SGE, SLURM), Multi-GPU Training, Large-scale Data Processing.

Professional Development

- Workshop on "Creating High-Quality Images for Publications," Lucy Family Institute for Data & Society, University of Notre Dame, Notre Dame, IN, USA, 2022.
- Workshop on "Machine Learning and Human-Computer Interaction," University of Asia Pacific, Dhaka, Bangladesh, 2019.
- Workshop on "Research Methodology and Thesis Writing," Uttara University, Dhaka, Bangladesh, 2019.

Leadership & Volunteering

- Social Chair, Bangladeshi Student Association, University of Notre Dame, 2023–2024.
- President, CSE Student Association, University of Asia Pacific, 2016–2017.
- Volunteer, IEEE 20th International Conference on Computer and Information Technology (ICCIT), Dhaka, Bangladesh, 2017.
- Volunteer, ACM International Collegiate Programming Contest (ACM-ICPC), Dhaka, Bangladesh, 2016 and 2017.